

Exact Belief-State Computation at Scale: Point-Based Recursion and Survivable-Set Compression

A computational companion to the belief-state safety-value theory

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

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Abstract

The belief-state safety-value theory is carried in exact rational arithmetic, but its published audits stop at instances a pencil could hold. This paper scales the exact programme to a medium instance — 48 augmented states (12 stock levels against the floor $\times$ 4 hidden-parameter cells), five controls — with two compression disciplines and no floating point anywhere. First, exact point-based evaluation: the alpha-set backup evaluated at probed beliefs returns the exact value at every probed belief, verified against definition-level trajectory enumeration at all probed points (seven rational beliefs, three horizons). Second, survivable-set compression: the belief kernel is stored as the antichain of maximal jointly survivable parameter sets rather than the full subset lattice — 48 stored sets in place of 180 raw subset evaluations across the twelve levels. The instance’s structure is carried completely: the indefinite-horizon kernel is singleton at the floor’s edge, the four coordinate pairs above it, and never larger — no policy keeps all four parameter cells safe, an obstruction proved by an averaging identity (every control’s drifts sum to $-\frac{2}{5}$ per step over the four cells) and certified at 60 steps for every three-periodic policy. The observation ladder is strict: blind values stabilize at $\frac{1}{4}$ (floor edge) and $\frac{1}{2}$ (everywhere above), one probe of the first parameter reaches 1 for every level above the edge ($\frac{1}{2}$ at the edge), and full observation is 1 throughout — one well-chosen probe exhausts the value of observation above the floor’s edge. All computations are exact integer and rational arithmetic, standard library only.

Keywords: exact computation; partially observable control; point-based value iteration; zero-suppressed sets; survivable sets; observation design

1. Introduction

The belief-state safety value $V_k(b)$ is exactly computable over rational data, but the audited demonstrations in the companion theory (hereafter the theory paper) use grids of at most 16 states. This paper asks whether the exact discipline survives scaling by an order of magnitude, and what new structure appears when it does. The answer is yes, with two working disciplines: *point-based evaluation*, which restores exactness at probed beliefs without enumer-

ating the simplex, and *survivable-set compression*, which replaces subset lattices by their maximal antichains. The medium instance — 12 floor levels $\times$ 4 hidden-parameter cells, five controls — turns out to carry a sharp structural obstruction (no control keeps all four parameter cells safe, ever), a strict observation ladder, and a clean band structure for the kernel; all of it is carried exactly.

2. The instance

The state is (z, θ) with stock z on the grid $z_0 \in \{1.0, 1.1, \dots, 2.1\}$ and hidden parameter $\theta = (\theta_1, \theta_2) \in \{+1, -1\}^2$: 48 augmented cells. Controls are $u \in \{(1, 1), (-1, -1), (1, -1), (-1, 1), (0, 0)\}$, and the one-step drift is additive,

$$\dot{z} = d(u, \theta) = -\frac{1}{10} + \frac{1}{5}(u_1\theta_1 + u_2\theta_2),$$

with the floor $z \geq 1$ and the matched action $u = \theta$ rising at $+\frac{3}{10}$. The belief lives on the four parameter cells at the observed stock level. No observation noise is present: the difficulty is the unknown pair, not the sensor.

3. Exact point-based evaluation

Proposition 1 (exact point-based values). *For finite A and fixed horizon k , the alpha-set backup of the theory paper’s Proposition (rational alpha-vectors) evaluated at a belief b — $\max_u \sum_{\theta} b(\theta) \mathbb{K}[\text{the cell survives the appended action}]$ — returns the exact value $V_k^\Pi(b)$ whenever the enumerated class element is optimal at b ; more generally the evaluated maximum over all class elements is the exact value by definition, with the attaining element as witness. No belief-space discretization error exists: the “point-based” step selects where to evaluate, not what to compute, and every returned number is an exact rational.*

On the instance, seven probed rational beliefs (and the uniform belief at every level) are evaluated exactly at horizons $k \in \{1, 3, 6\}$, each against definition-level trajectory enumeration over all action sequences — exact equality throughout (Section 6.). The finite-horizon table over the twelve levels is the paper’s first output: $V_6(b_0) = \frac{1}{4}$ at the floor’s edge, $\frac{1}{2}$ on 1.1 through 1.5, and 1 from 1.6 up (the

doomed cells survive six steps from higher stock), monotone nonincreasing in the horizon at every level.

4. Survivable-set compression

Proposition 2 (kernel by antichain). *A parameter-cell set S is jointly survivable if one (possibly time-varying) blind policy keeps every cell in S above the floor indefinitely. The indefinite-horizon belief kernel is $\{b : \max_S b(S) = 1\}$ over the maximal jointly survivable sets, and the stored object is the antichain of maximal sets (componentwise order), not the subset lattice: on the instance, 48 stored sets in place of 180 raw subset evaluations over the twelve levels. The bands are: the four singletons at $z_0 = 1.0$; the four coordinate pairs $\{++,+-\}$, $\{++,-+\}$, $\{-+,-+\}$, $\{-+,-+\}$ — at every level $z_0 \geq 1.1$; nothing larger, ever.*

Two structural facts explain the bands. First, the coordinate pairs rise: under the alternation $(1,1) \leftrightarrow (1,-1)$ both cells of $\{++,+-\}$ drift $+\frac{3}{10}, -\frac{1}{10}$ per pair of steps, net $+\frac{1}{10}$ with dips of $\frac{1}{10}$, safe forever from $z_0 \geq 1.1$. Second, nothing larger survives:

Proposition 3 (the averaging obstruction). *Every control's drifts over the four parameter cells sum to $-\frac{2}{5}$ per step. Hence over any time window the four cells' cumulative displacements sum to a negative number, so some cell exits the floor band: no policy — blind or otherwise, of any horizon — keeps all four parameter cells safe indefinitely. With one cell sacrificed, the search certifies at 60 steps that every three-periodic policy keeps at most a coordinate pair; the pairs are therefore maximal, and the indefinite-horizon kernel at the uniform belief is $\frac{1}{2}$ for every $z_0 \geq 1.1$ and $\frac{1}{4}$ at the edge.*

The obstruction is structural, not numeric: it is the same “someone must pay for the drift” identity that the timing layer of the calculus prices, here read across parameter cells rather than time.

5. The observation ladder

Proposition 4 (one probe exhausts observation above the edge). *Reveal θ_1 only (one probe; the support becomes a coordinate pair). The post-probe support is jointly survivable at every $z_0 \geq 1.1$, so the one-probe value is 1 there; at the floor's edge it is $\frac{1}{2}$: the controller keeps exactly one cell of the pair alive, and the probe's expected value is half the single-cell mass. Fully observed play (matched action, drift $+\frac{3}{10}$) is 1 everywhere. The ladder per level is therefore*

$$\frac{1}{4} < \frac{1}{2} < 1 \quad (z_0 = 1.0), \quad \frac{1}{2} < 1 = 1 \quad (z_0 \geq 1.1) :$$

above the floor's edge, a single well-chosen probe exhausts the value of observation; the second parameter is worth nothing once the first is known, because the surviving pairs share a coordinate.

6. Verification methods

Every number is regenerated by one deterministic script in exact integer and rational arithmetic, standard library only: the finite-horizon table by definition-level enumeration over all action sequences at $k \in \{1, 2, 3, 6\}$ and all

twelve levels; the maximal-set classification by two- and three-periodic policy search with 60-step certificates, maximality by antichain filtering; the averaging obstruction by the identity $10d(u, \theta) = -1 + 2(u_1\theta_1 + u_2\theta_2)$ summed over the cells; the ladder by post-probe support enumeration; the point-based claims by exact evaluation at seven probed rational beliefs against the definition path; the compression census (48 stored sets, 180 raw) by counting; and every headline number asserted present in this text.

7. Delimitations

The instance has deterministic per-cell dynamics and error-free probes; observation noise enters the companion theory's stochastic layer, not this one. Point-based evaluation is exact at the probed beliefs and does not approximate between them. The compression is exact for the stored objects and makes no claim beyond the antichain's size. Scaling further (hundreds of cells, many parameters) is the next step, for which the antichain representation — not the subset lattice — is the argued data structure.

8. Conclusion

The exact discipline scales, and scaling pays in structure: a medium instance exhibits an averaging obstruction no policy can beat, a two-band kernel, and an observation ladder in which one probe is everything. The two working disciplines — evaluate exactly where you probe, store the antichain not the lattice — are the transferable result.

References

- Abaee, A. An obstruction calculus for viability under incomplete observation (submitted).
Abaee, A. Probabilistic sufficiency for the obstruction calculus: belief-state safety values under partial observation (companion theory paper, same repository).

Declarations

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_exact_belief_computation_v1_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all values, sets, bands, and census counts verbatim.

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